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Intelligent Assessment Method of Structural Reliability Driven by Carrying Capacity Sustainable Target: Taking Bearing Capacity as Criterion

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Abstract: Large-scale building structures are subject to numerous uncertain loads during their service life, leading to a decrease in structural reliability. Real-time analysis and accurate prediction of structural reliability is a key step to improve the bearing capacity of buildings. This study proposes an intelligent assessment method for structural reliability driven by a sustainability target, which incorporated digital twin technology to establish an intelligent evaluation framework for structural reliability. Under the guidance of the evaluation framework, the establishment method of a structural high-fidelity twin model is formed. The mechanical properties and reliability analysis mechanism are established based on the high-fidelity twin model. The theoretical method was validated by experimental analysis of a rigid cable truss construction. The results showed the simulation accuracy of the high-fidelity twin model formed by the modeling method is up to 95%. With the guidance of the proposed evaluation method, the mechanical response of the structure under different load cases was accurately analyzed, and the coupling relationship between component failure and reliability indicators was obtained. The twinning model can be used to analyze the reliability of the structure in real time and help to set maintenance measures of structural safety. By analyzing the bearing capacity and reliability index of the structure, the safety of the structure under load is guaranteed. The sustainability of structural performance is achieved during the normal service period of the structure. The proposed reliability assessment method provides a new approach to improving the sustainability of building bearing capacity.

Keywords: sustainable structure; digital twin; mechanical response; reliability analysis; experimental validation

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1. Introduction

Large public buildings use spatial and prestressed structures [1] that are exposed to wind loads and temperature effects during their service life, significantly affecting their serviceability and bearing capacity [2]. The frequent collapse of structures poses a serious threat to life and property. The collapse of the World Trade Center in 2001 [3] and the unexpected collapse of a terminal building at Paris Charles de Gaulle Airport in 2004 were examples of these events. Therefore, reliability analysis of structures has consistently been a crucial area of research in civil engineering [4].

Regarding reliability analysis of structures, most existing studies use finite element simulation and experimental validation [5]. Lu et al. [6] studied the failure mechanism of a 6 m-span plane trusses string structure through experiments and numerical simulations, analyzing the effects of different cable rupture types on structural reliability. Vaezzadeh et al. [7] discovered the progressive collapse mode of cable-net structures through experimental analysis, with the main factor being constraint failure. Ozer et al. [8] conducted

modal analysis of a bridge structure by establishing a finite element model (FEM), comparing frequency and vibration mode changes to judge reliability. Bojorquez et al. [9] evaluated the impact of corrosion on the reliability of reinforced concrete structures using Monte Carlo simulation and stochastic modeling methods, highlighting its important role in structural failure and the necessity for consideration in reliability design.

The rapid development of smart buildings and intelligent construction has put forward new requirements for the bearing capacity of structures [10]. Against this backdrop, improving the sustainability of building bearing capacity is an urgent issue [11]. Traditional reliability analysis methods cannot achieve continuous evaluation and rely on data generated by experiments. Real-time analysis and accurate prediction of structural reliability are significant for improving the sustainability of buildings. Digital twin (DT) technology plays a role in the interactive mapping of deficiency and excess and spatiotemporal evolution analysis in structural health monitoring [12]. DT technology realizes the interactive mapping between structural entities and virtual models. The establishment of the twin model of the structure provides a powerful tool for the intelligent evaluation of reliability. Driven by the twin model, the sustainability of structural bearing capacity can be effectively improved. [13].

Wang et al. [14] proposed a damage identification method for cable dome structures by combining DT and deep learning. This method can identify damage type, location, and degree intelligently with high accuracy and strong robustness. Ritto et al. [15] explored an ensemble method based on finite element modeling and machine learning under the twin concept. This method effectively integrates engineering reality with virtual models, offering reliable decisions for structural safety control. Zhu et al. [16] proposed a DT-based safety assessment method for dams. This assessment method makes automated and efficient monitoring of dam conditions possible. Liu et al. [17] proposed a DT-based intelligent evaluation method in order to enhance the constructional safety degree of prestressed steel structures. This method ensures a safety level for structures and elevates the intelligent level of construction safety evaluation.

Driven by sustainability goals and the DT concept, this study proposes an intelligent assessment method for structural reliability. The safety of the structure under load is ensured by analyzing the bearing capacity and reliability index of the structure. The sustainability of structural performance is achieved during the normal service period of the structure. DT uses a high-fidelity dynamic virtual model to simulate the state and behavior of physical entities. As a link between the real physical world and the virtual digital space, it is a key enabling technology for structural safety assessment. By establishing an HF twin model to map the mechanical state of the structure in real time, the method accurately evaluates the failure mode and reliability development of the structure. The method provides decision support for structural bearing capacity maintenance. Integrating DTs into structural reliability evaluation can allow analysis of the mechanical response under extreme operating conditions in the twin model, reducing the cost of experimental research. More importantly, this research method realizes real-time analysis and accurate prediction of structural reliability, effectively improving the sustainability of structural bearing capacity. The organizational structure of this paper is as follows.

The second section puts forward the theoretical method of reliable evaluation. In this chapter, the framework and process of evaluation are formed. Driven by the evaluation process, the establishment method of the twin model is first formed. Based on the establishment of the twin model, the mechanical properties and reliability of the structure are analyzed. Analyzing the mechanical properties and reliability of the structure provides a basis for improving the sustainability of the carrying capacity.

Based on the theoretical method in Section 2, Section 3 is verified by experiments. During testing, a high-fidelity virtual model of the structure was first established. The simulation ability is improved by modifying the model. The unfavorable load conditions and failure modes are obtained in the multi-span and multi-stage loading process. Finally, the reliability index of the structure is analyzed in different extreme load conditions.

Section 4 is the conclusion of this study.

2. Methodology

To lift the sustainability of building bearing capacity, this study proposes an intelligent reliability assessment method for structures. By using DT technology, an HF twin model of structures was created, which enables analysis of structural mechanical properties and reliability.

2.1. Intelligent Evaluation Framework Driven by DTs

During normal service life, a structure is affected by external factors such as temperature and wind loads, which threaten its safety. During the lifetime of the structure, internal causes such as component relaxation and component corrosion will also affect its reliability. Real-time analysis and accurate prediction of structural reliability have become essential scientific issues. The physical parameters, geometric parameters and boundary parameters which represent the stiffness and mass of the structure are all time-varying functions [18], making the analysis of structural reliability a multidimensional mechanical problem that couples time and space [19]. DTs, driven by multidimensional virtual models and integrated data, can monitor, simulate, predict, optimize, and meet practical functional services and application requirements through closed-loop interactions between the virtual and real worlds [20]. DTs encompass information technologies such as the IoT and AI, fully considering the spatiotemporal evolution characteristics of structural mechanical states, and realizing HF mapping of the structure. Multidimensional, multiple process, multiple parallel operation time simulations, and integrated management of deficiency and excess are achieved under the support of the five-dimensional space of DTs. The structural reliability intelligent evaluation framework driven by DTs is shown in Figure 1. Establishing an HF twin model based on the state of the actual structure is the first step to achieve reliability assessment. Comparing the measured values of structural mechanical properties with the simulated values, the basic parameters of the structure in the model are modified to improve the simulation ability of the model. Based on the establishment of the virtual model, the mechanical properties of the structure are analyzed. In the virtual model, the mechanical response corresponding to the applied load of different spans is analyzed to obtain unfavorable load conditions. The failure mode of the structure is analyzed under the unfavorable load condition. For extreme load conditions of the structure, such as component failure, the reliability index of the structure is analyzed in the virtual model. The residual bearing capacity and reliability of the structure are obtained by analyzing the mechanical properties of the structure, which provides a basis for ensuring the safety of the structure.

Under the guidance of DT, this study achieves intelligent evaluation of structural reliability by considering the spatiotemporal evolution of the structural system and its interaction with the virtual reality interaction. First, an HF twin model is established for interactive mapping in the same physical space, forming DT data for structural mechanical performance and reliability analysis. By integrating DT technology with the effects and mechanical performance parameters of the structure, a multiple-scale model of the longitudinal dimension of the structural system is realized. Based on the real-time evolution of the load during the service life of the structure, an operational system of structural dynamic collaboration driven by DT is established.

Virtual models were established according to the physical structure to accurately map the structural mechanical state. In this study, the structural mechanical performance was analyzed through the multi-span and multi-stage distribution of loads. Extreme operating conditions were set in the virtual model to analyze the structural failure probability and obtain the structural reliability index ultimately. Based on the analysis of mechanical performance and reliability, maintenance strategies were developed to effectively improve the sustainability of structural bearing capacity.

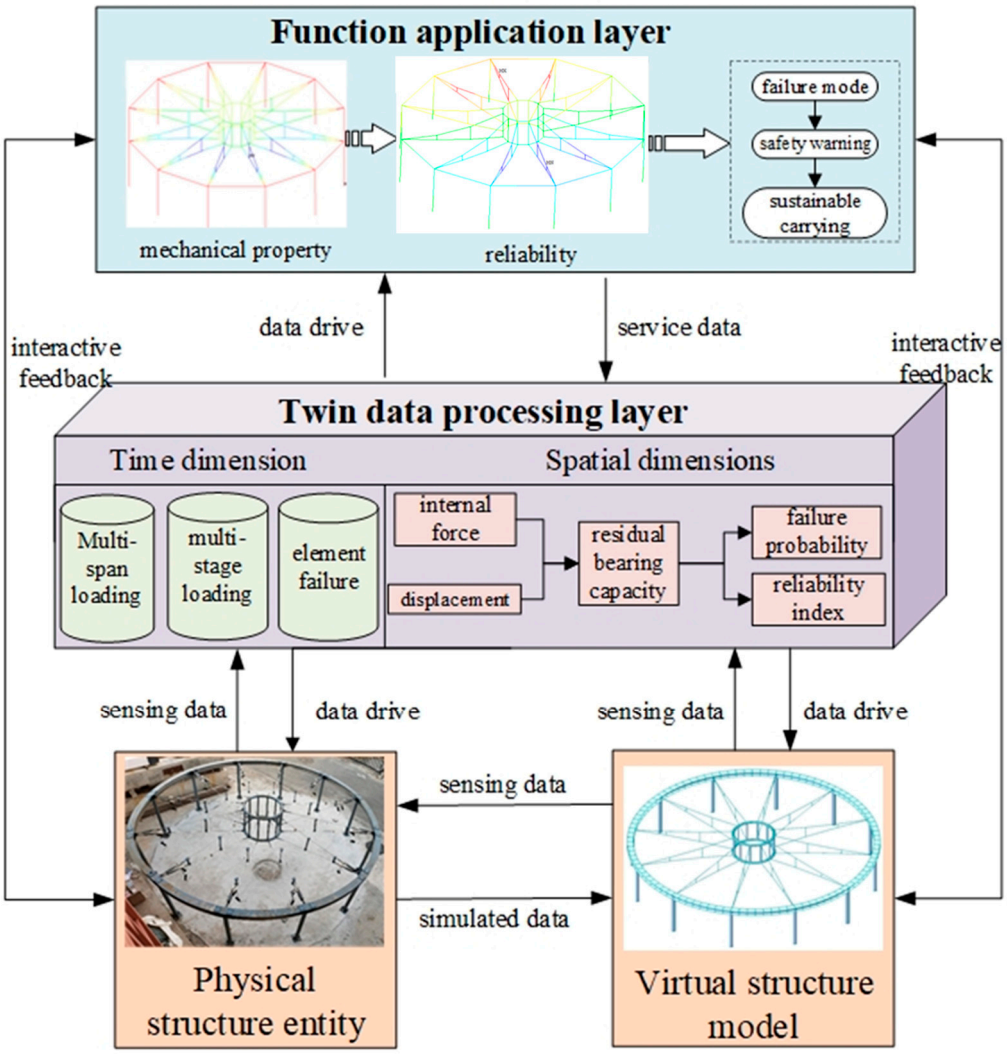


Figure 1. The structural reliability intelligent evaluation framework driven by DTs.

2.2. Establishment of the DT Model

Establishing a structural HF twin model is the first step in achieving intelligent reliability assessment. This section analyzes the traditional finite element modeling process and forms a model correction method. Following the establishment of the DT model, the structural reliability can be analyzed in real time and the development trend of the reliability index under extreme operating conditions can be accurately predicted. With the DT model established, it effectively reduces the dependence on experimental data.

2.2.1. Construction of FEM

Through the study of structural performance and reliability, the traditional simulation method relies on FEMs [21]. The FEM analysis software ANSYS is used to analyze the structural performance, with the following modeling process.

(1) Assign material properties and real constants to the corresponding components.
(2) Arrange key joints in the software, establish corresponding component units, and form a structural model.
(3) Apply corresponding constraints to the connection joints of the structure.
(4) Apply the loads that the structure bears at the designated joints.
(5) Obtain the structural mechanical response under the action of the loads.

2.2.2. Modification of Virtual Model

On the basis of establishing the FEM, this study proposes a physical property optimization method based on a genetic algorithm [22,23]. In the optimization process, physical attributes such as component size are used as optimization parameters. The lower and upper limits are determined according to the machining and construction accuracy of physical properties. The error ($\Delta\mu$) between the displacement simulation value of the structure in its formed shape and the measured value is taken as the optimization objective. The optimization process of attribute parameters in the virtual model is expressed as Equation (1).

$$\begin{cases} a_1 \le \delta_1 \le b_1 \\ a_2 \le \delta_2 \le b_2 \\ \cdots \\ a_n \le \delta_n \le b_n \end{cases} \xrightarrow{\text{iteration}} \min(\Delta\mu) \xrightarrow{select} \begin{cases} \delta_1^{opt} \\ \delta_2^{opt} \\ \cdots \\ \delta_n^{opt} \end{cases} \quad (1)$$

where δ_i equals the ith optimization variable, such as the prestress of a component, and a_i and b_i denote the lower and upper limits of the ith optimization variable. The optimization algorithm is used to adjust the basic parameters of the model to minimize the error between the simulated and measured vertical displacement values, while δ_i^{opt} is the optimal design parameter corresponding to the minimum error. By modifying the structural physical properties in the model, the accuracy of model simulation is significantly improved. In this model, the mechanical state of the structure is analyzed in real time and the safety risk is predicted, which provides support for the intelligent evaluation of structural mechanical performance and reliability. The basic parameter optimization process of the virtual model is shown in Figure 2. The parameters are optimized in Python. The minimum error between the simulated value and the measured value is taken as the objective function, and the value range of each basic parameter (size, prestress) is taken as the constraint condition. Driven by the genetic algorithm, the parameters are optimized to improve the simulation ability of the virtual model.

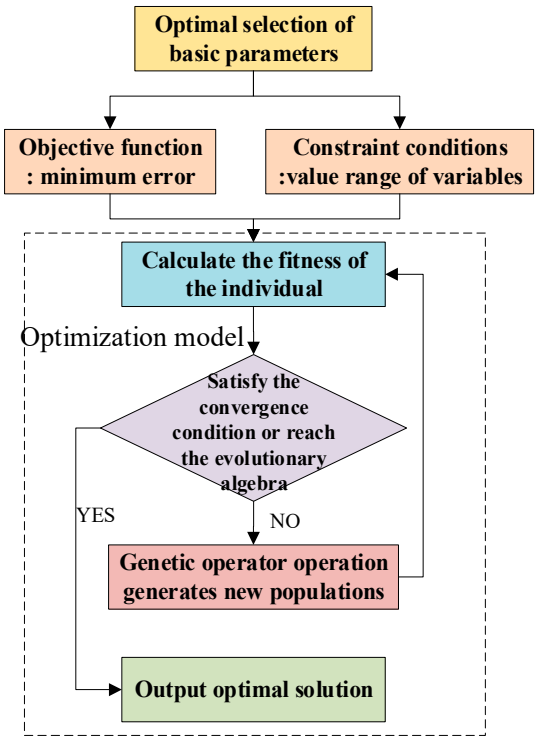


Figure 2. Model modification process based on the genetic algorithm.

2.3. Analysis of Structural Mechanical Performance and Reliability

For the analysis of structural performance under different load cases, this study developed a twin model of the experimental model formed by the theoretical method in Section 2.1. In the formed shape, the HF of the twin model is verified by comparing the measured and simulated values of the structural mechanical parameters. Driven by the genetic algorithm, the basic parameters in the model are optimized to improve the simulation ability of the model. A variety of static load conditions are arranged in the test model and the twin model to obtain the mechanical response of the structure. By setting multi-span loads, the most unfavorable load on the structural performance is obtained. Multi-stage loading is performed for the most unfavorable load to obtain the failure mode of the structure. The reliability analysis of the structure is carried out in the twin model for extreme loads, such as component failure. By analyzing the mechanical properties and reliability of the structure, the bearing capacity of the structure is evaluated to improve the sustainability of the structure. This process is shown in Figure 3. In the analysis of structural mechanical performance and reliability, this study used a tension meter and a total station to capture the changes in cable force and displacement. The twin model of the structure is established by ANSYS.

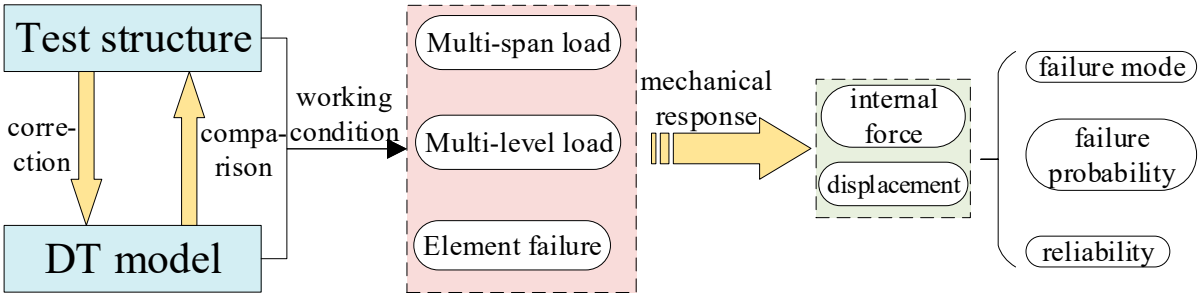


Figure 3. Analysis process of structural mechanical performance and reliability.

This study conducted an analysis of the internal forces and displacements of the structure under multi-stage loading to evaluate the failure mode of the structure. The loads corresponding to the ultimate limit state (ULS) and serviceability limit state (SLS) of the structure were obtained. The reliability analysis of the structure after component failure was carried out in the twin model. Response surface methodology and the Monte Carlo method were applied in the reliability analysis of the structure. Firstly, the response equation was obtained using the response surface method, and then 1 million sampling simulations were performed with Latin hypercube sampling (LHS) to obtain the structural failure probability.

In the DT model, the effect of structural failure on the ULS and SLS was studied. For the ULS, the loading condition (LS) is expressed in Equation (2) [24], and the function of the failure is expressed in Equation (3).

$$L(ULS) = 1.3 \times dead\ load + 1.5 \times live\ load \quad (2)$$

$$f(ULS) = [\sigma] - \sigma_{max} \quad (3)$$

where $[\sigma]$ represents the limit value of the component stress, which is equal to the material strength divided by the safety factor, σ_{max} is the maximum stress of the components in the structure, and $f(ULS) < 0$ refers to structural failure.

For the SLS, the LS is expressed in Equation (4) [24], and the failure function is expressed in Equation (5).

$$L(SLS) = 1.0 \times dead\ load + 1.0 \times live\ load \quad (4)$$

$$g(SLS) = [\mu] - \mu_{max} \quad (5)$$

where $[\mu]$ represents the limit value of the structural displacement, which is equal to the structural span divided by the safety factor, μ_{max} represents the maximum displacement of the structure, and $g(SLS) < 0$ represents structural failure. By analyzing the structural failure function, the structural failure probability was obtained, and the development trend of the structural reliability index was captured. Based on the structural mechanical performance and reliability analysis, this study provides a basis for the safety maintenance of the structure, effectively ensuring the sustainability of the structural bearing capacity.

3. Experimental Verification

To demonstrate the feasibility of the research method, an experimental verification was carried out on a rigid cable truss construction. In the experimental model, the span of the cable truss was 6 m. The model consists of 12 bays of cable truss, inner ring beam, outer ring beam, and outer column. The radial cables include upper radial cables (URCs) and lower radial cables (LRCs) with a diameter of 9 mm, and the cable forces of the URCs and LRCs in formed shape are 15 kN and 16 kN, respectively. The single-bay cable truss is connected to the radial cable by two struts, the inner ring beam includes upper and lower ring beams, which are connected by inner struts. The height of the inner strut is 0.75 m, and the inner radius is 0.5 m. The height of the outer column is 1.5 m. The experimental model of this study is shown in Figure 4. The mechanical properties of cable members are the focus of this study. Tension meters are arranged in each cable to collect the change of cable force. The displacement of each node under self-weight condition is collected by total station. By comparing the measured and simulated values of the displacement, the size and prestress of the component are corrected. The modified virtual model can accurately map the mechanical properties of the structure. The feasibility of the model is verified by comparing the measured and simulated values of the cable force. On the basis of establishing a high-fidelity twin model, the state of the structure under various load conditions can be accurately analyzed in the model. By analyzing the mechanical properties of the structure under extreme conditions, the reliability of the structure is judged, which provides a basis for improving the bearing capacity of the structure. The structure is formed by adjusting the sleeve in the test model. The vertical load of the joint is applied by arranging weight blocks on the joint.

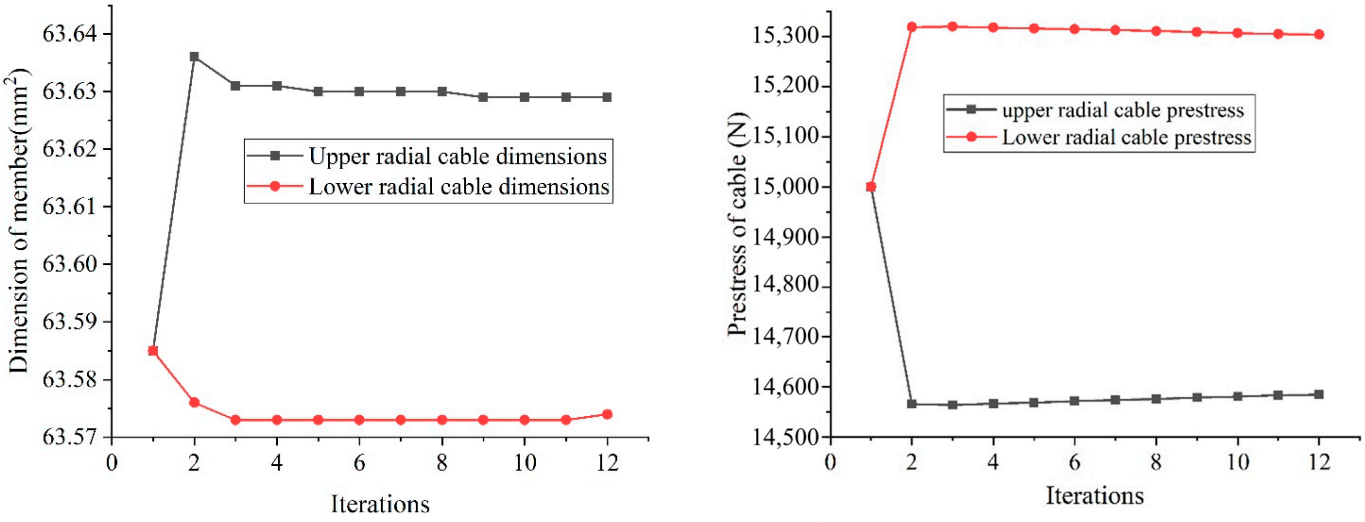


Figure 4. The experimental model: (① URC, ② LRC, ③ middle strut, ④ outer strut, ⑤ upper ring beam, ⑥ lower ring beam, ⑦ inner strut, ⑧ outer ring beam, ⑨ outer column, ⑩ tension meter, ⑪ collecting box).

3.1. Structural Twin Modeling

Establishing an HF twin model is the first step of intelligent assessment. Based on the modeling method in Section 2.2, this study first established the FEM of the structure. The cable components in the structure are built using Link10 elements, while the rigid components are built using Beam188 elements. The boundary conditions for the cable components are based on three-direction articulation. The base of the structure's columns is a rigid constraint. The cable elements are prestressed by applying initial strain. The vertical load is set in the model to simulate the actual structure loading process.

Based on the FEM, the structural vertical displacement in its formed shape was obtained. The error between the simulation and measured values of the vertical displacement was taken as the optimization objective. The physical properties optimized were the dimensions and prestress of the cables. The optimization parameters were iterated under the drive of a genetic algorithm, as shown in Figure 5. The optimal physical properties were obtained after 12 iterations. The error between the obtained vertical displacement in the twin model and the measured value was within 5%. In the process of virtual model optimization, the values of key parameters of genetic algorithm are shown in Table 1.



(**a**) The process of dimension iterations (**b**) The process of prestress iterations

Figure 5. The iteration process of physical properties.

Table 1. Optimization parameters of the model.

Parameter	Adjustable Range	Value	Meaning
η	[0, 1]	0.01	learning rate
γ	[0, ∞]	500	genetic algebra
ς	[0, ∞]	50	population size

Based on the optimized physical properties, an HF structural twin model was established, as shown in Figure 6. To verify the effectiveness of the twin model, the simulation and measured values of the radial cable forces were obtained for the structure in its formed shape. The comparison showed that the fidelity of the cable force analysis by the DT model was also within 5%. The comparison of the cable force simulation and measured values is shown in Table 2. Since the structure is symmetric in geometry and loading, Table 2 shows the comparison of the radial cables of a single bay.

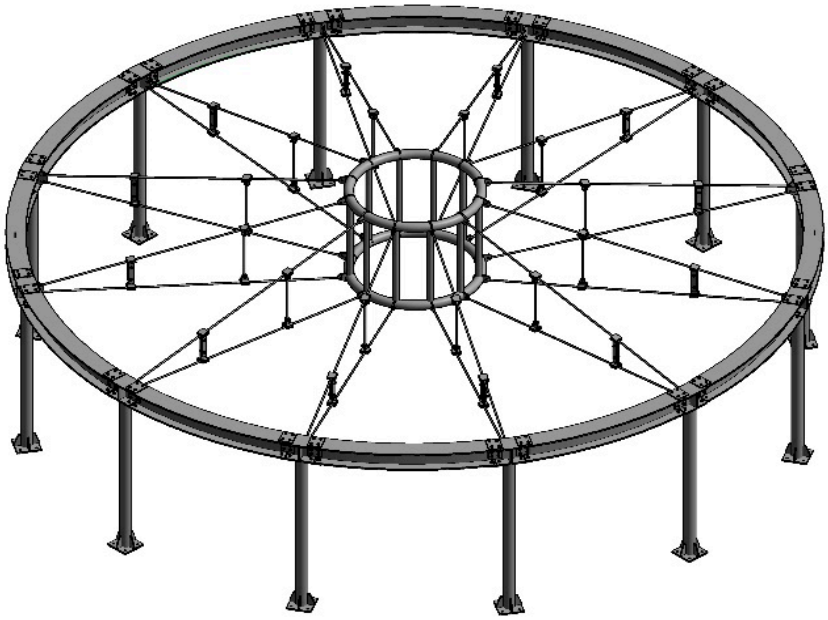


Figure 6. The HF twin model of the structure.

Table 2. The comparison of cable force between simulation value and measured value.

Component No.	Measured Value (N)	Simulation Value (N)	Error Value
URC 1	12,821	13,424	4.7%
LRC 1	14,588	14,034	−3.8%

3.2. Mechanical Response of Obtained Structure Based on DTs

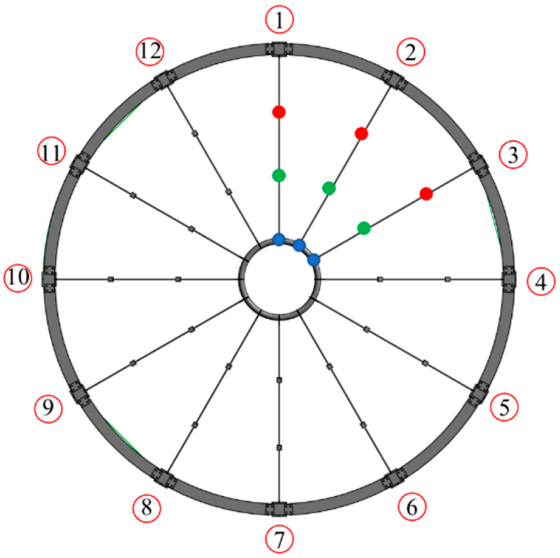
A DT model can accurately and effectively simulate the mechanical properties of the results. This section analyzes the mechanical response of the structure under different span loads using the DT model. The structure's failure modes were obtained through multi-stage loading according to the most unfavorable load cases. The structural reliability index was obtained for extreme operating conditions.

3.2.1. Mechanical Performance of Different Load Conditions

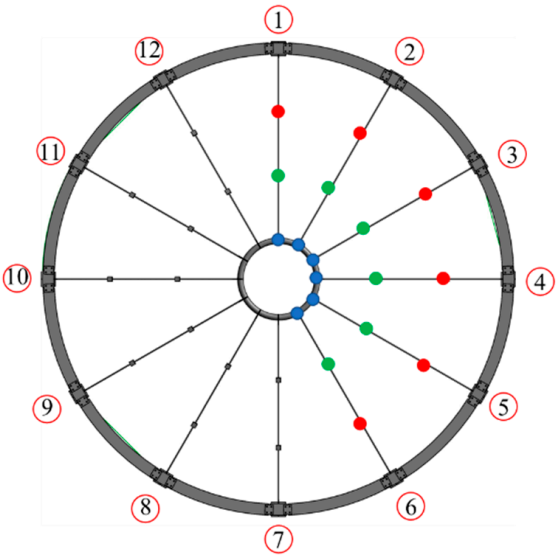
Four load cases were analyzed in this study in static load analysis, as shown in Table 3. According to the literature [25], the constant load was taken as 0.15 kN/m^2, and the surface load was converted to an equivalent joint load [26], where the joint load on the inner strut was 33 N, on the middle strut was 92.3 N, and on the outer strut was 146 N. The specific LSs are shown in Figure 7.

Table 3. The load cases.

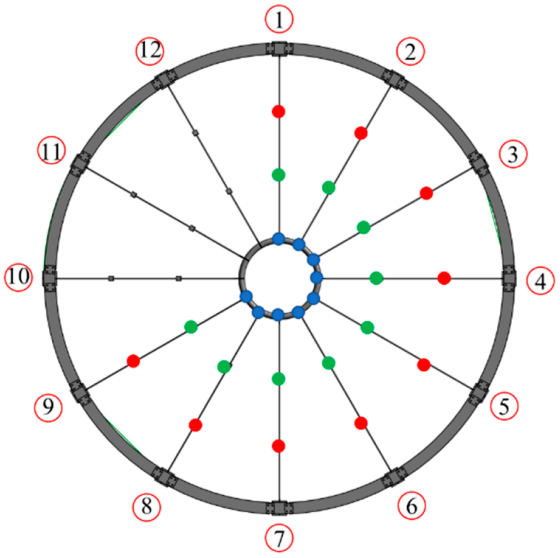
	Conditions of Load Combination
LS 1	1/4-span constant loads
LS 2	1/2-span constant loads
LS 3	3/4-span constant loads
LS 4	Full span constant loads



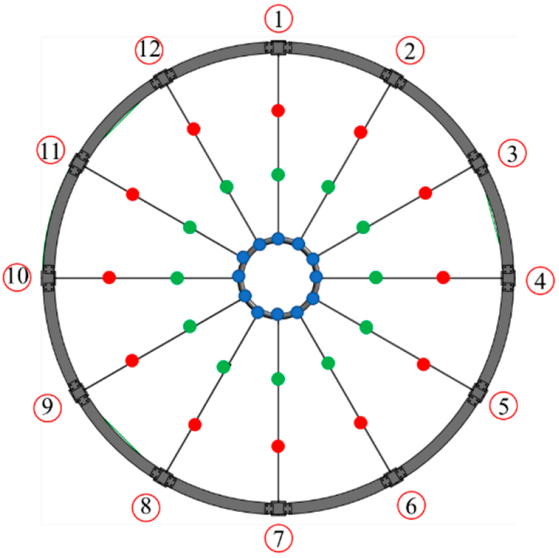
(**a**) 1/4-span constant loads



(**b**) 1/2-span constant loads



(**c**) 3/4-span constant loads

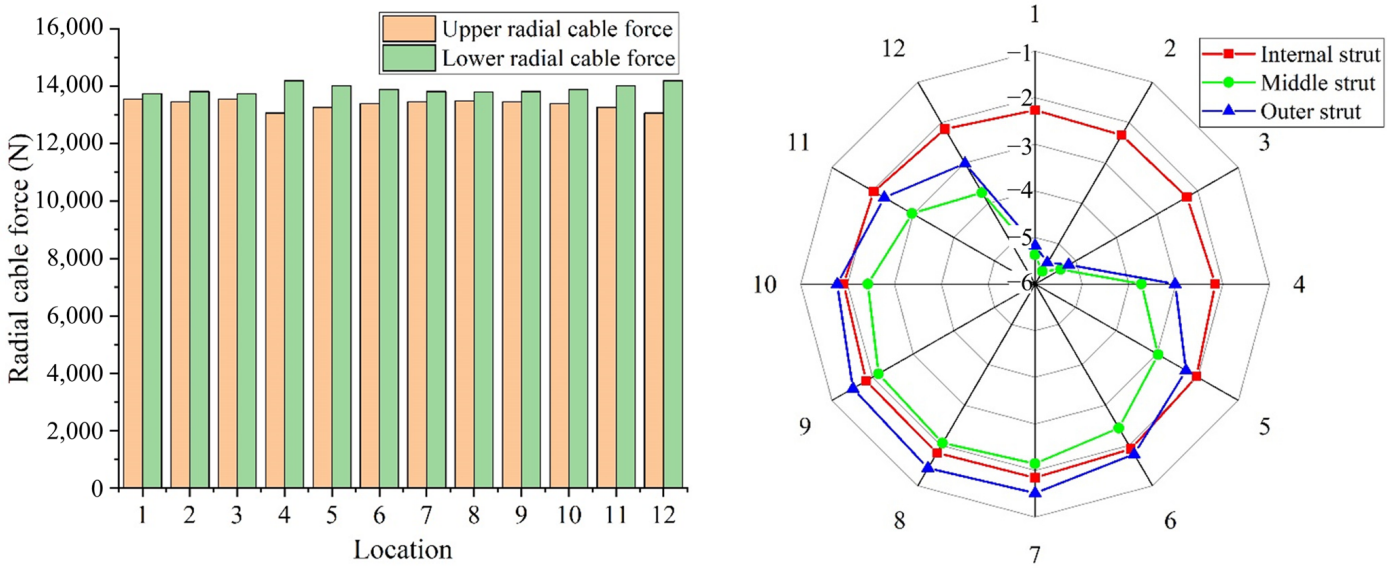


(**d**) Full span constant loads

Figure 7. The specific LSs (red is 146 N, green is 93.2 N, and blue is 33 N, The number in the figure indicates the axis position of the cable truss).

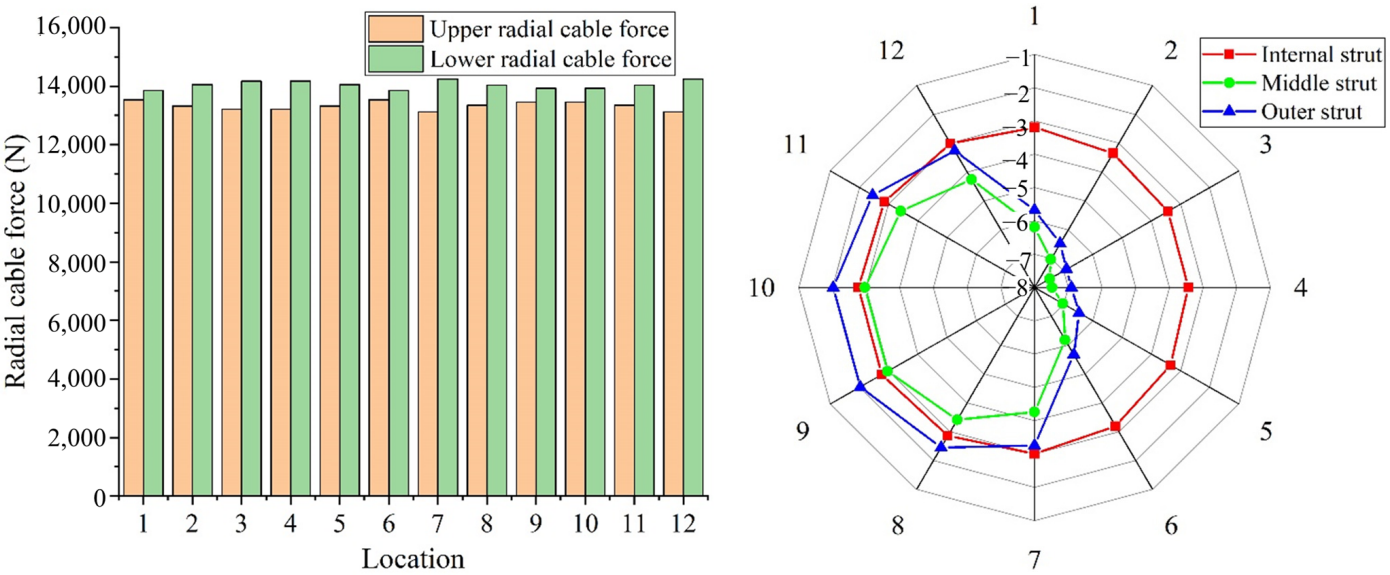
(1) Quarter-span constant loads

Quarter-span loads were applied to the first, second, and third bays of radial cables in the experimental model to load the structure. The DT model's simulation analysis of internal forces and displacements is shown in Figure 8. The vertical coordinate axis in Figures 8, 9, 10 and 11b represents the vertical displacement, which gradually decreases from inside to outside. Under the action of load, the displacement of the external strut is maximal, so the corresponding curve is on the inside.



(**a**) Radial cable forces (**b**) Joint displacements

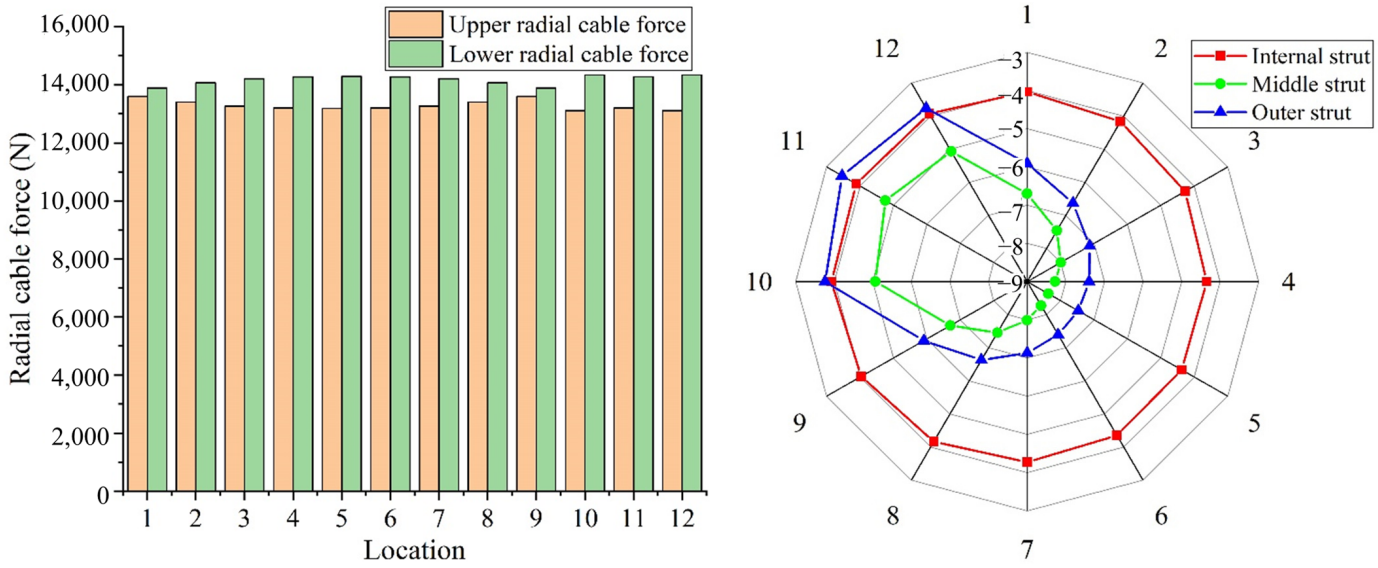
Figure 8. Simulation analysis of the mechanical properties of the structure.



(**a**) Radial cable forces (**b**) Joint displacements

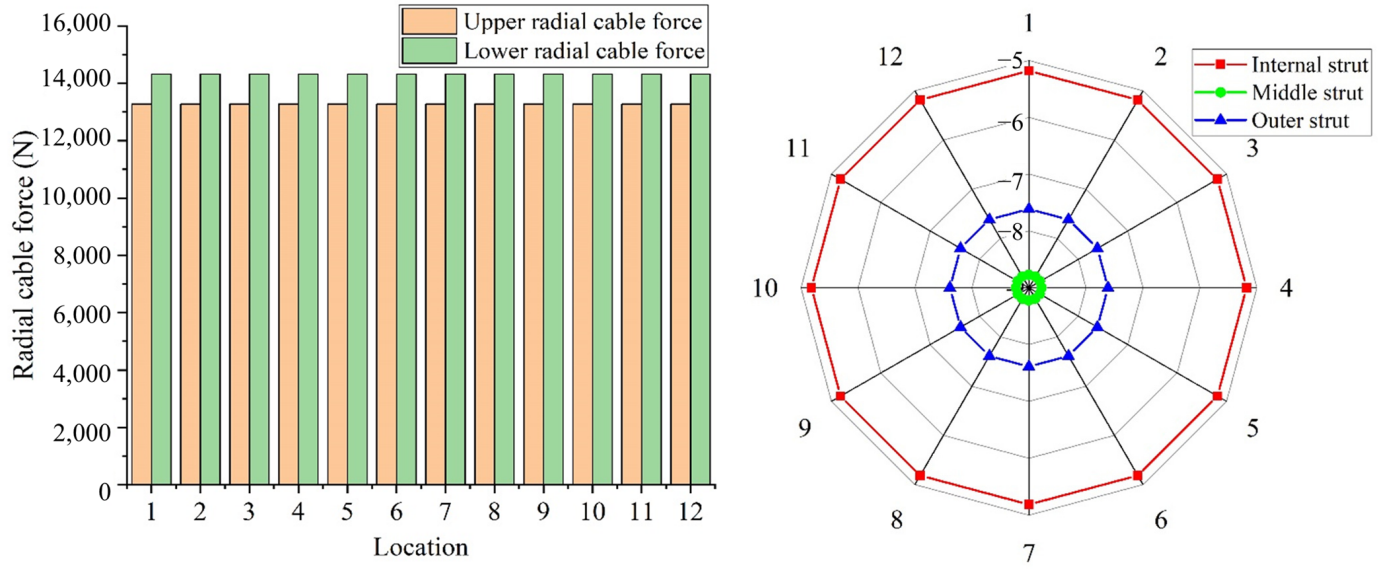
Figure 9. Simulation analysis of the mechanical properties of the structure.

Compared with the structure in formed shape, when a 1/4-span load was applied, the force in the URC increased, while that in the LRC decreased. The force in the URC of the first and third bays increased the most, increasing from 13,424 N to 13,555 N, a variation rate of 1%. The force in the LRC of the first and third bays decreased the most, decreasing from 14,034 N to 13,745 N, a variation rate of 2.1%. The vertical displacement at the loading point changed the most, and the joint displacement of the middle strut was the largest. When a 1/4-span load was applied, the joint displacement of the middle strut in the second bay was the largest, at −5.68 mm.



(**a**) Radial cable forces (**b**) Joint displacements

Figure 10. Simulation analysis of the mechanical properties of the structure.



(**a**) Radial cable forces (**b**) Joint displacements

Figure 11. Simulation analysis of the mechanical properties of the structure.

(2) Half-span constant loads

One to six bays of radial cables in the test model were loaded to apply a 1/2-span load to the structure. The internal forces and displacements obtained from the twin-model simulation analysis are shown in Figure 9.

When a 1/2-span load was applied, the maximum force of the URC for the first and sixth bays was 13,540 N, while the maximum force of the LRC for the third and fourth bays was 14,168 N. Compared with the formed shape, the upward cable force of the third and fourth bays changed the most, with a change rate of 1.6%. The LRC force of the seventh and twelfth bays changed the most, with a change rate of 1.5%. The vertical displacement at the loading position changed the most, and the joint displacement of the middle strut was the largest, reaching −7.48 mm when a 1/2-span load was applied to the third and fourth bays.

(3) Three-quarter-span constant loads

Bays 1–9 of radial cables in the test model were loaded to apply a 3/4-span load to the structure. The internal forces and displacements obtained from the twin-model simulation analysis are shown in Figure 10.

When a 3/4-span load was applied, the maximum force of the URC for the first and ninth bays was 13,599 N, while the maximum force of the LRC for the fifth bay was 14,291 N. Compared with the formed shape, the upward cable force of the tenth and twelfth bays changed the most, with a change rate of 2.3%. The LRC force of the tenth and twelfth bays changed the most, with a change rate of 2.2%. The vertical displacement at the loading position changed the most, and the joint displacement of the middle strut was the largest, reaching −8.36 mm when a 3/4-span load was applied to the fifth bay.

(4) Full-span constant loads

When subjected to full-span constant load, the internal forces and displacements of the test model were analyzed, and the results are shown in Figure 11 using the twin-model simulation.

When the full-span load was applied, the force on the URC was 13,271 N, and that on the LRC was 14,326 N. The force on the URC decreased by 1.1% as a whole, while the force on the LRC increased by 2.1% as a whole. The maximum joint displacement of the middle strut was −8.78 mm, and the minimum joint displacement of the inner strut was −5.18 mm.

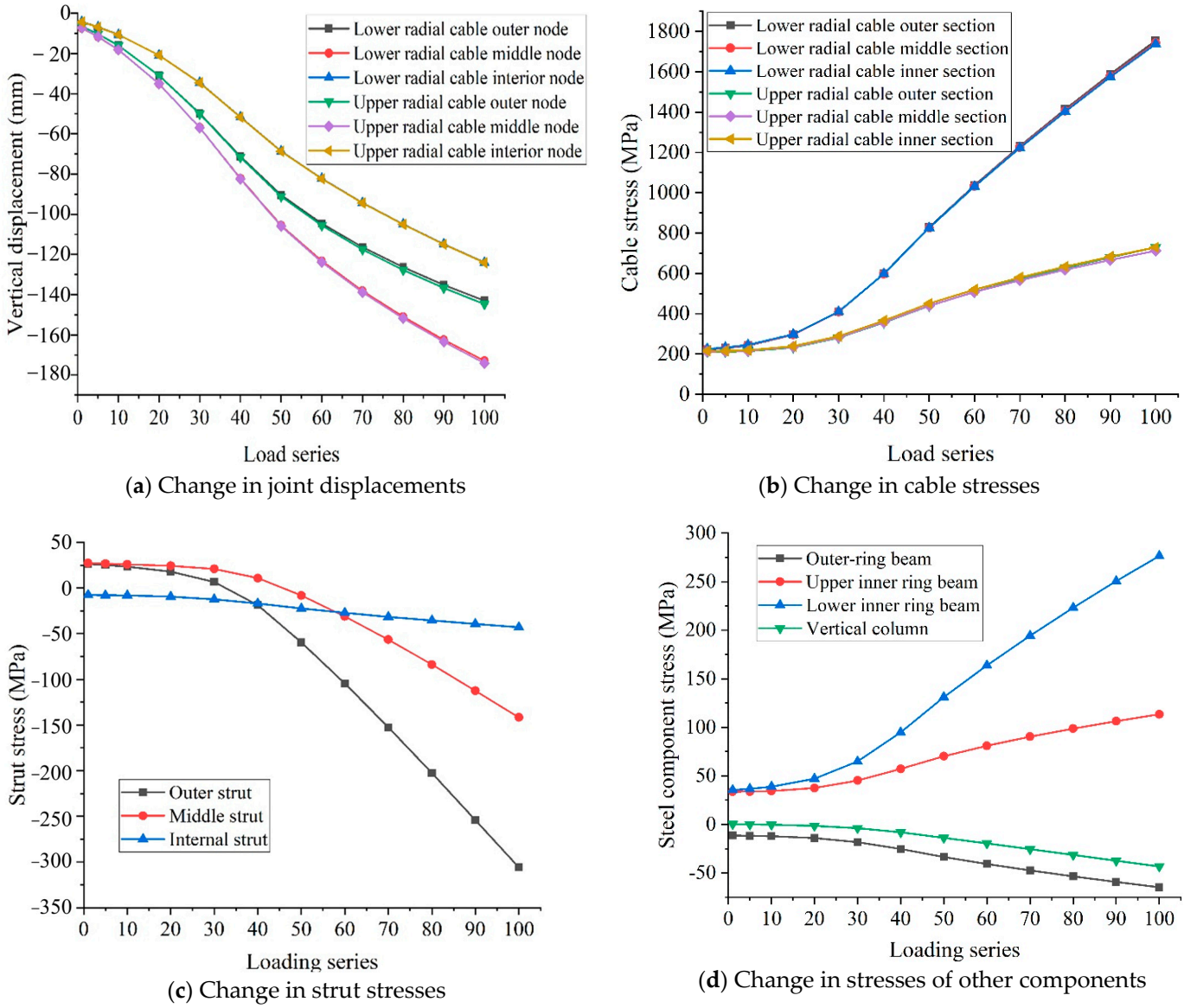
3.2.2. Structural Failure Mode

By analyzing the structural mechanical performance under different span LSs, it was found that the maximum internal force and displacement of the structure occurred under full-span constant load. The failure modes of the cable truss structure are transfinite deformation, cable breaking, strut destabilization, and cable relaxation [27,28]. In the twin model, multi-level loading with full-span constant load was carried out to analyze the trend of changes in the structure's vertical displacement and internal forces. The mechanical response of the structure as the load increases is shown in Figure 12. Under the full-span load, the displacement of the upper and lower joints of the same strut is the same (Figure 12a). The same cable has the same stress in all sections (Figure 12b).

According to the cable structure technical specification [29–33], the limit value of vertical displacement under normal service conditions is 1/250 of the span, that is, 24 mm. In this experiment, transfinite deformation occurred when loading level 15 constant load. The tensile strength of the cable and the strength of the steel component used in the experiment were 1670 MPa and 345 MPa, respectively. As shown in Figure 12, when loading up to level 95 constant load, the LRC force first reaches the breaking force. Among the struts, the stress change in the outer strut is the most obvious. Through the analysis of the twin model, the failure mode of the structure's ultimate bearing capacity state is identified as cable breaking.

3.2.3. Reliability of Components after Failure

In order to improve the sustainability of structural bearing capacity, a reliability analysis of the structure was conducted. In this paper, the Monte Carlo method based on response surface was used to analyze the damage situation of the cable truss structure. In this method, the structural response surface is first fitted using the sampling method, and then the response surface equation is used to replace the structure's FEM. After that, the structure is analyzed for reliability using the Monte Carlo method.



(**a**) Change in joint displacements

(**b**) Change in cable stresses

(**c**) Change in strut stresses

(**d**) Change in stresses of other components

Figure 12. Structural mechanical response of multi-stage loading process.

Structural failure probability under different conditions was calculated using Equations (3) and (5). The displacement limit value was set to 24 mm. According to the analysis in Section 3.2.2, cable breaking is the failure mode of the ULS. In Equation (5), the limit value of the component stress is the cable's tensile strength divided by the safety factor. According to the cable structure technical specification [29–33], the safety factor is 2.0. The extreme value of stress is 835 MPa. Under the guidance of the theoretical method in Section 2.3, this study conducted an evaluation of the structural reliability under four typical operating conditions [34–36], as shown in Table 4.

As shown in Table 4, the failure of the LRC has the greatest impact on the structural reliability, and the reliability index of the SLS is close to 0. Therefore, during the structural service life, the stress state of the LRC should be closely monitored to prevent component relaxation or excessive stress. In addition to typical operating conditions, the structural reliability can be analyzed in real time for other operating conditions using the DT model. Structural bearing capacity maintenance can be carried out based on the changes in the reliability index.

Table 4. Evaluation of structural reliability.

Components Failure	SLS		ULS	
	Failure Probability	**Reliability Index**	**Failure Probability**	**Reliability Index**
Single URC failure	3.71887×10^{-1}	0.3269	7.59028×10^{-3}	2.4280
Single LRC failure	4.26797×10^{-1}	0.1845	8.49569×10^{-3}	2.3869
Single middle strut failure	3.13185×10^{-1}	0.4868	6.14143×10^{-3}	2.5039
Single outer strut failure	2.81372×10^{-1}	0.5788	5.79581×10^{-3}	2.5243

3.3. Discussion

Using the twin-modeling method, an HF twin model of the test structure was established. The simulation accuracy of the twin model was found to be over 95% through comparison, which fully demonstrates the feasibility of the DT modeling method.

The structural mechanical performance was analyzed during multi-span loading. It was found that the URC on the edge span had the maximum cable force, and the LRC on the middle span had the greatest change in cable force. The joint displacement of the middle strut was greater than that of the outer and inner struts, and the joint displacement was the largest for the loading on the middle span. The sensitive areas of the structure were determined by analyzing its response to multi-span loading.

The structural failure modes and structural bearing capacity were obtained through multi-stage loading. The structure exceeded the deformation limit when subjected to a constant load at level 15. When loaded to level 95, the LRC force reached its breaking point. Under the ultimate load, the cable force of the lower radial cable is 2.2 times that of the upper radial cable. Therefore, the stress performance of the lower radial cable should be emphasized in the loading process. Through the analysis of multi-stage loading, it is found that grade 15 constant load corresponds to the normal service limit state of the structure, and grade 95 constant load corresponds to the ultimate state of the structure's bearing capacity.

The importance of various components was obtained by analyzing the structure's reliability. The order of importance was LRC > URC > outer strut > middle strut. In the DT model, structural reliability under various conditions can be analyzed accurately, providing the basis for improving the sustainability of structural bearing capacity. In the normal service limit state, the reliability index of the failure structure of a single lower radial cable is close to 0, which is the most unfavorable extreme condition of the structure. When a single member fails, the reliability index corresponding to the lower radial cable is a third of that of the outer strut.

4. Conclusions

This study proposes an intelligent reliability assessment method for improving the sustainable carrying capacity of the structure by integrating DT technology. Through the establishment of an HF twin model, real-time analysis and accurate prediction of structural reliability were achieved. On the basis of analyzing the mechanical performance and reliability of the structure, strategy support was provided to ensure the sustainability of the bearing capacity of the structure. The main conclusions are as follows.

(1) The establishment method of the HF twin model was formed by combining the genetic algorithm and finite element modeling theory. By comparing the simulated and measured values of structural mechanical parameters, the physical properties of the model are modified. The DT model can accurately map the structural mechanical state with an error within 5%. Additionally, the DT model can analyze the structural mechanical response under various operating conditions in real time, reducing the dependence on experimental data.

(2) Using the DT model, the analysis method of the structural mechanical performance and reliability was formed. The DT model can accurately analyze the structural carrying capacity and the development trend of reliability indicators. The structural safety status can be accurately judged based on the reliability index, which provides the basis for improving the sustainability of structural bearing capacity.

(3) During the experiment, the advantages of the proposed intelligent assessment method were verified. Under the guidance of the theoretical method, the mechanical behavior law of the structure under multi-span load arrangement is obtained, and the key stress nodes and adverse conditions are captured. By analyzing the development trend of displacement and internal force under multi-stage loads, the ultimate bearing capacity of the structure is obtained. By reaching component failure, the reliability index and key components are obtained by simulation in the twin model. Real-time analysis of the structural mechanical performance and reliability effectively ensured the safety of the structure, providing ideas for improving the sustainable bearing capacity of the structure.

This study focused on the reliability and bearing capacity of cable structures. It provides ideas for other types of structural safety assessment. By obtaining adverse working conditions, failure modes, and reliability indices in the twin model, it provides a basis for improving the sustainability of structural bearing capacity. For concrete structure and masonry structure, it is necessary to further study the influence of material ratio and cooperative work between materials on structural reliability. The large volume of existing buildings necessitates higher requirements for structural health monitoring. It is important to improve the sustainability of existing building carrying capacity by updating the twin model with the dynamic and static mechanical data of the fusion structure driven by twin data and realizing accurate identification of structural damage. Driven by the sustainability goal of structural carrying capacity, it is advised that future studies focus on forming efficient maintenance measures based on the reliability index of structures. Taking into account the whole life cycle of the building, low energy consumption in the operation and maintenance of the structure is also a future research direction.

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Nomenclature

Formula symbol	physical meaning
DT	digital twin
HF	high fidelity
F_{DT}	evaluation framework driven by DTs
L_{PSE}	physical structure entity
L_{VSM}	virtual structure model
L_{TDP}	twin data processing layer
L_{FA}	functional application layer
C_N	connections between each component
FEM	finite element model
$\Delta\mu$	displacement error
δ_i	ith optimization variable
a_i	lower limits of the ith optimization variable
b_i	upper limits of the ith optimization variable
δ_i^{opt}	optimal design parameter corresponding to the minimum error
LHS	Latin hypercube sampling
ULS	ultimate limit state
SLS	serviceability limit state
LS	loading condition
$[\sigma]$	limit value of the component stress
σ_{max}	maximum stress of the components in the structure
$[\mu]$	limit value of the structural displacement
μ_{max}	maximum displacement of the structure
URC	upper radial cable
LRC	lower radial cable

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